



The NIVA Level Controller MS1 is the ideal solution to control liquid levels with limited switching space, e.g. in:

- Pump stations
- Wells
- Pump chambers



The NIVA Level Controller MS1 is engineered especially for use in sewage works and pumping stations with liquids heavily charged with solid matter such as raw sewage etc. Thanks to the good chemical and thermal properties our level controllers are resistant to lees, uric acid, fecal sewage water, oils, petrol, diesel oil, emulsions, alcohol, fruit acids, and even many chemicals. For use at temperatures of up to 80 °C (176 °F).

Available versions:

Type	Cable	Cable length (approx.)	Order Number
SPDT	TPR/PVC 3 x 0,75 mm ²	5m (16ft)	40 000105
SPDT	TPR/PVC 3 x 0,75 mm ²	10m (30ft)	40 000110
SPDT	TPR/PVC 3 x 0,75 mm ²	20m (65ft)	40 000120
SPDT	TPR/PVC 3 x 0,75 mm ²	30m (98ft)	40 000130

Other cable types and lengths are available upon request

Application:

For use in municipal, industrial, commercial and domestic applications.

Electronic connection:

Connection of level controller	Wire		
	gray	black	brown
For emptying a tank normally open	insulate	X	X
For filling a tank normally closed	X	insulate	X
Alarm high level	insulate	X	X
Alarm low level	X	insulate	X

Technical data subject to change

Technical data:

Specific weight:	0.95 - 1.05 or according to specification
Max. temperature:	80°C
Breaking capacity:	1mA / 4V - 5A / 250V*
Switch point:	10°
Protective system:	IP 68 / 7 bar
Equipment group:	II
Cable cross section:	3x0.75 mm ²
Height / diameter:	180 / 100 mm
Housing quality:	Polypropylene (PP)
Housing Color:	Orange
Cable quality:	TPR/PVC
Cable color:	Orange

* Micro-switch with gold-plated contacts made especially for low currents in electronic circuits

Level Regulator MS1



Nominal Values:

Specific weight	0.95-1.0 depending on model
max. Operating temp.	80° C
Power	1mA / 4 V – 5A / 250 V
Switching angle	10°
Protection type:	IP68 / 2 bar
Protection class	II
Housing	Polypropylene (PP)
Height/diameter	190 mm / 100 mm
Cable	TPR3x0,75mm²

Resistance List:

Raw sewage, faecal discharges, slurry, household sewage, discharge water from washing machines, baths and showers, detergents, emulsions containing petrol, diesel oils, fats, oil, acids etc.
rain water, groundwater, chlorinated water, saltwater
brine, vegetable oils, fruit acids, alcohol

EC Declaration of Conformity

In accordance with
EC Low-Voltage Directive 2006 / 95 / EG
EC Environmental Protection Directive RoHS 2002 / 95 / EG

NOLTA GmbH, 35091 Coelbe,
We,

hereby declare that the products manufactured by us comply with the current fundamental requirements with regard to safety and environmental protection as outlined in the EC directives with regard to their design and construction and this also applies to the version brought to the market by us. This declaration becomes null and void if alterations are made to the level regulator without being agreed with us beforehand.

Applicable harmonised standards:

- DIN EN 60730-1 (VDE 0631-1)
- DIN EN 60730-2-15 (VDE 0631-2-15)
- DIN IEC 60730-1 (VDE 0631-1)

Coelbe, 28/10/2013

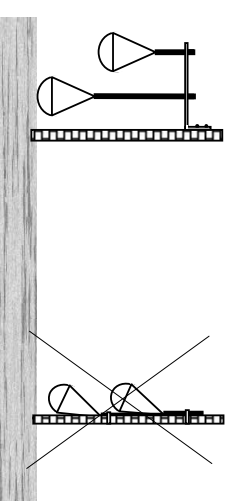
Dr.-Ing. Jochen Knake

Managing Director

Wolfgang Seip

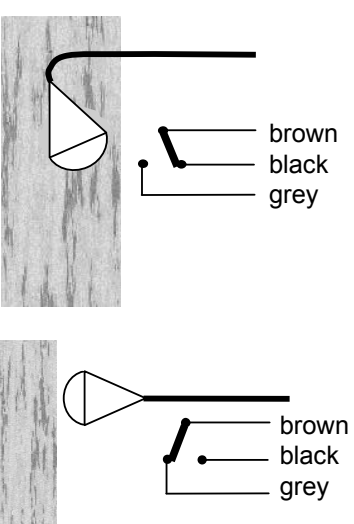
Quality Management

Assembly:



Electrical Connection:

Before connecting to electricity, rectifying faults or carrying out maintenance work, the electrical system must be disconnected from the power supply. Ensure that neither water nor moisture can enter via the cable ends.



To fill: connect brown – grey
To empty: connect brown – black
The unused wire must be insulated

Maintenance:

If the float switch is installed and assembled correctly, it should work practically maintenance-free for many years. Depending on the level of contamination in the medium, the system only needs to be checked from time to time and the float switch cleaned if necessary.

Nolta GmbH
Industriestr. 8
35091 Coelbe
Germany

Tel: +49(0)6421/9859 0
Fax: +49(0)6421/9859 28
www.nolta.de
info@nolta.de



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Level Controller MS1



Manual and Safety Instructions

Nolta GmbH
Industriestr. 8
35091 Cölbe
Germany

Ph +49(0)6421/9859 0
Fax +49(0)6421/9859 28
www.nolta.de
info@nolta.de

General Information



Read this manual before installing and activating this product.
Respect all safety instructions and local laws and regulations.



The installation may only be executed by qualified electricians.
This product may only be used according to its intended use set forth in this manual.

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The following Symbols and hazard statements are used in this installation, operating and safety instructions:

Hazard statements

Danger



Indicates a hazardous situation which, if not avoided, will result in death or serious personal injury.

Warning



Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.

Caution



Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.

Notes



A blue or grey circle with a white graphical symbol indicates that an action must be taken.



A red or grey circle with a diagonal bar, possibly with a black graphical symbol, indicates that an action must not be taken or must be stopped.



If these instructions are not observed, it may result in malfunction or damage to the equipment.

The hazard statements are structured in the following way:

Signal Word

Description of Hazard

Consequence of ignoring the warning.
Action to avoid the hazard.

Technical Data

Technical Data

Specific weight:	0,95-1,05 g/cm ³
Housing quality/-color:	Polypropylene (PP) orange
Cable quality/-color:	Thermoplastic rubber (TPR) orange
Cable cross-section/D:	3x0,75mm ² (SPDT) D=6,5mm (40 0001..) 3x0,75mm ² (NO) D=6,5mm (40 0701..) 3x1,0mm ² (SPDT) D= 7,6mm (40 0011..)
Power:	1mA/5VDC-5A/250VAC
Protective rating:	IP68 / 7bar
Protective class:	II
Switching angle:	Approx. 10°
Tmax:	80°C (occasionally)
Dimensions (HxD):	Approx. 170mm / 100mm
Confirmations/Certificates:	CE, EAC

Table 4

Intended use:

The level controller MS1 is especially suitable for use in wastewater treatment plants and pumping stations with solid matter contaminated liquids such as raw sewage. Thanks to the good chemical properties, this level regulator is resistant to suds, fecal sewage, fruit acids, alcohols as well as many weak chemical or organic emulsions in an environment up to 80°C.

Installation

Assembly:

The controllers must be installed freely suspended with a min. of 15 cm tether length. (See „Figure 2“) For a correct assembly we recommend using the separately available accessory "Cable Brackets" (40 02).

Every MS1 generates a single switching Point. To define the required water level, install one controller at the low and one at the high switch point. Connect them to the corresponding connectors in your control box in accordance with the diagram in „Table 1“.

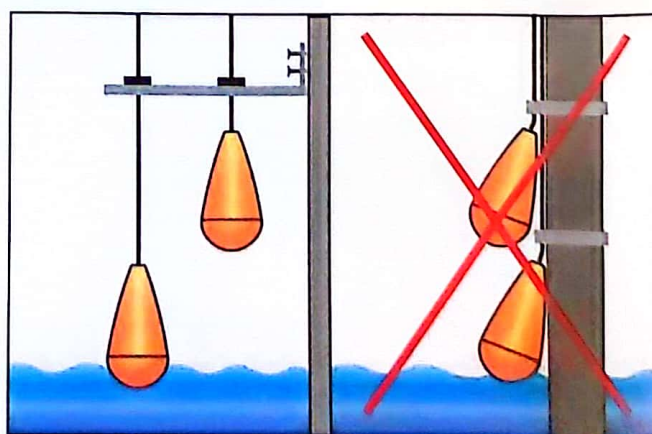


Figure 4

You may define more switch points with additional controllers in the same manner. For example "Start pump 2" or a high or low alarm.

Electrical Connection



Danger Electric Shock

Before connecting to electricity, rectifying faults or carrying out maintenance work, the electrical system must be disconnected from the power supply.

Ensure that neither water nor moisture can enter at the cable ends.

Always insulate unused cable ends.



The electrical connection must be carried out by an authorized electrician in accordance with local laws and regulations.



The controllers are intended for connection with a control box and corresponding control voltages. The connection to a 230V power source is technically possible, however if reverted back to lower voltages, a reliable functionality cannot be guaranteed.

	Brown	Black	Grey
To empty (Normally open)	✓	✓	✗
To fill (Normally closed)	✓	✗	✓
High Level Alarm	✓	✓	✗
Low Level Alarm	✓	✗	✓

Table 3

✓ = Connect

✗ = Insulate

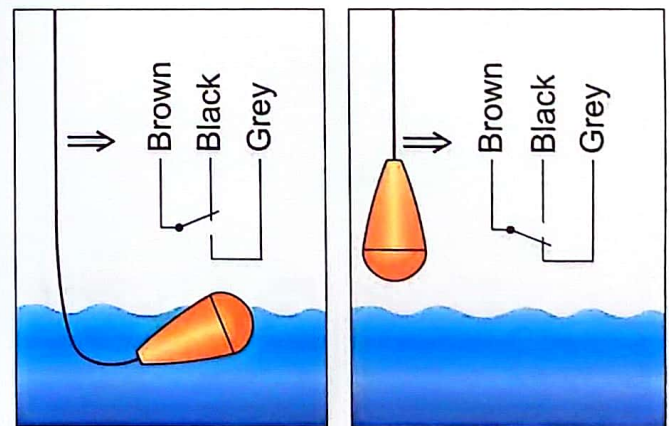


Figure 3

Maintenance and Disposal

Maintenance:

If the level controller is installed and assembled correctly, it should work practically maintenance-free for many years. Depending on the level of contamination in the medium, the system only needs to be checked from time to time and the controller cleaned if necessary.

Disposal:

This product or parts of it must be disposed of in an environmentally sound way: Use the public or private waste collection service. If this is not possible, please contact your NOLTA dealer.



NOLTA

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EU Declaration of Conformity

We hereby declare that the level controller specified below will, due to its design and construction, Comply with the relevant regulations listed.

Product Designation

MS1

Manufacturer

Nolta GmbH
Industriestr. 8
35091 Cölbe

EU directives / Harmonized standards /
national techn. Standards - Specifications

Low Voltage Directive 2014/35/EU

Restriction of use of certain hazardous substances in electrical and
electronic equipment (RoHS) -- Directive 2011/65/EU

EN 60730-1:2011

Automatic electrical control and regulating devices Part 1: General
requirements

EN 60730-2-15:2010

Automatic electrical controls and control devices for domestic use
and similar applications Part 2-15: Particular requirements for
automatic electrical air, water, and water level control devices

EN 61010-1:2010

Safety requirements for electrical equipment for measurement,
control and laboratory use -- Part 1: General requirements

Authorized representative
Name and address

David Loechelt
Nolta GmbH
Industriestr. 8
35091 Cölbe

We confirm that a CE mark according to the European directives is affixed to the above mentioned level
controller.

30.04.2018

Date

CEO
Dr. Ing J. Knake

Head of Quality Management
D. Loechelt